MATERIALS 2

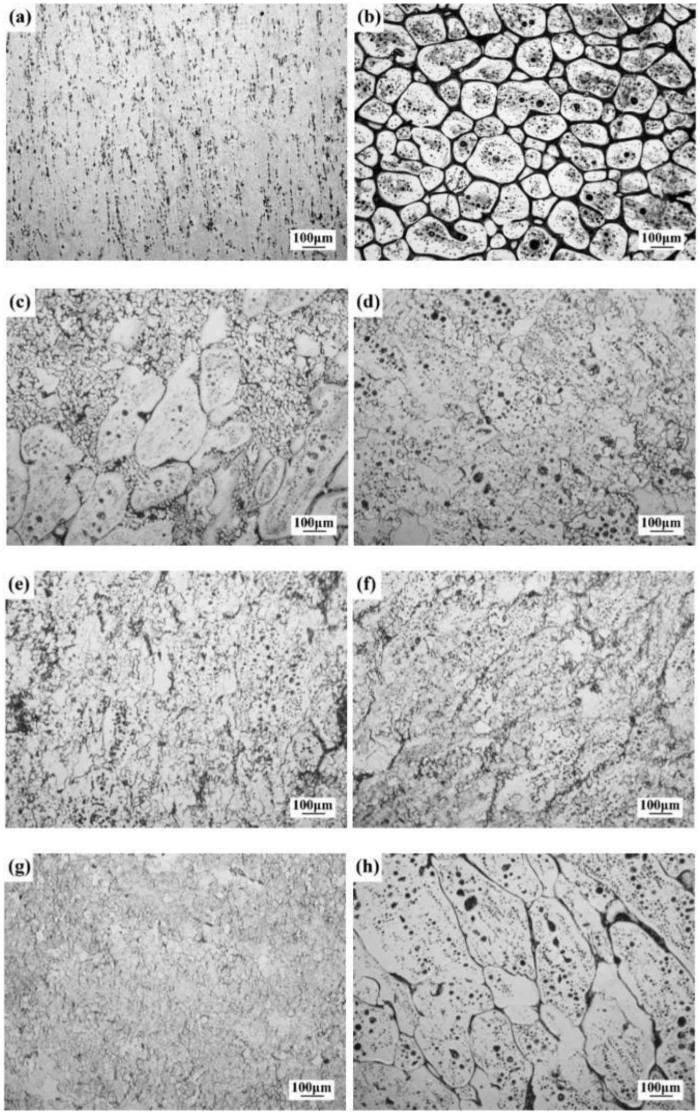
Lecture 2: Phase Diagrams I

Fida MAJDOUB, Ph.D.

Contents

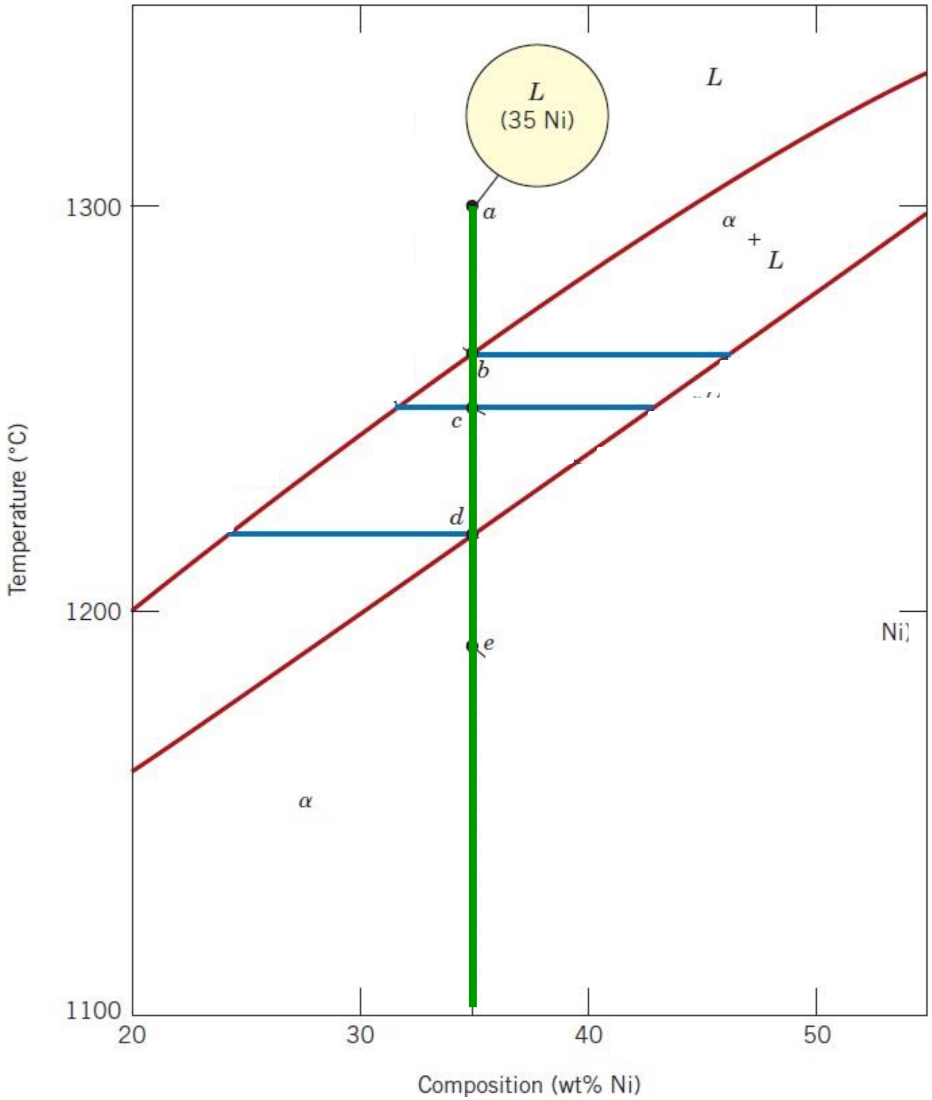
Microstructure

- A microstructure of an alloy describes the appearance of the internal of the structure in different length scales.
- A microstructure is defined by the type, structure, number, shape and topological arrangement of phases.
- A microstructure shows defects, impurities, grains, and grain boundary.
- A microstructure is observed using a range of microscopy techniques.
- A microstructure indicates the mechanical properties of an alloy.



Equilibrium Cooling

Slow Cooling of Copper-Nickel Alloy

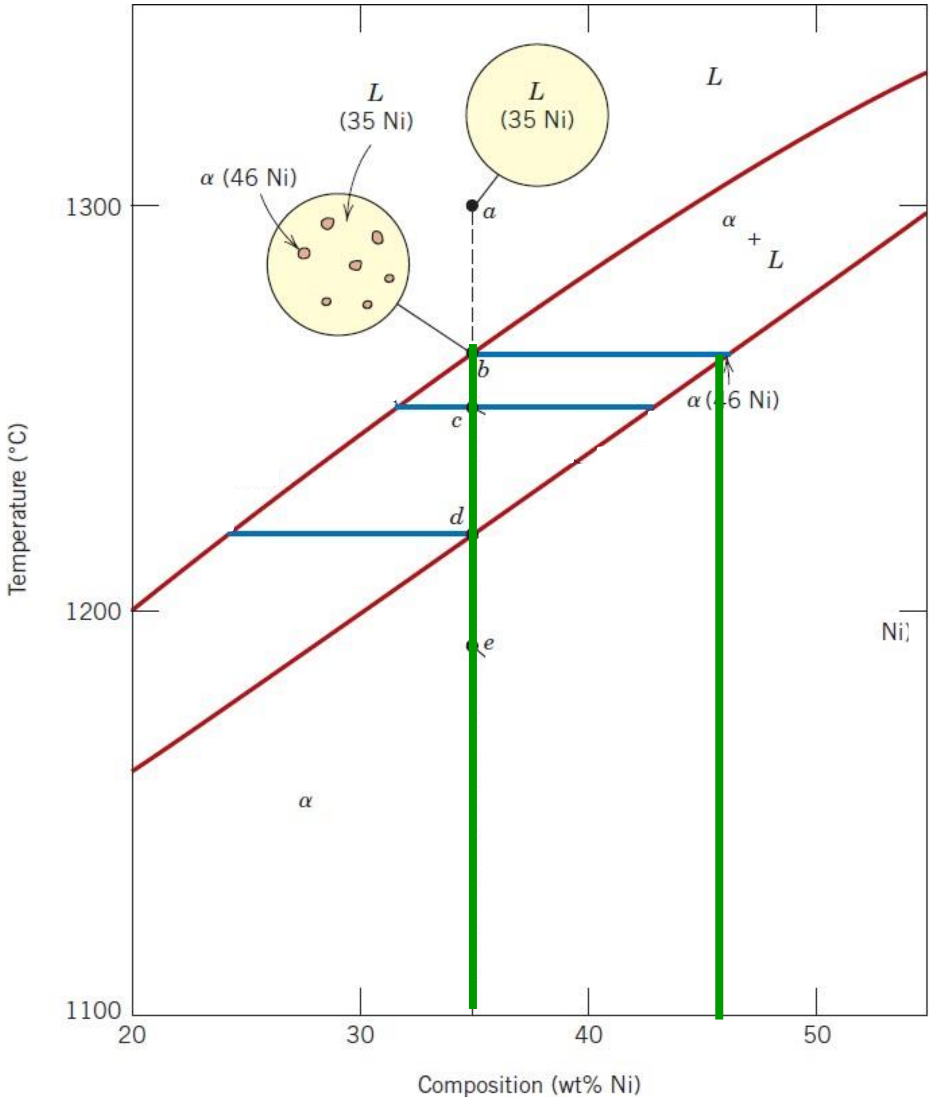


a:

Liquid phase: $35\ wt\%$ Ni and $65\ wt\%$ Cu

Equilibrium Cooling

Slow Cooling of Copper-Nickel Alloy

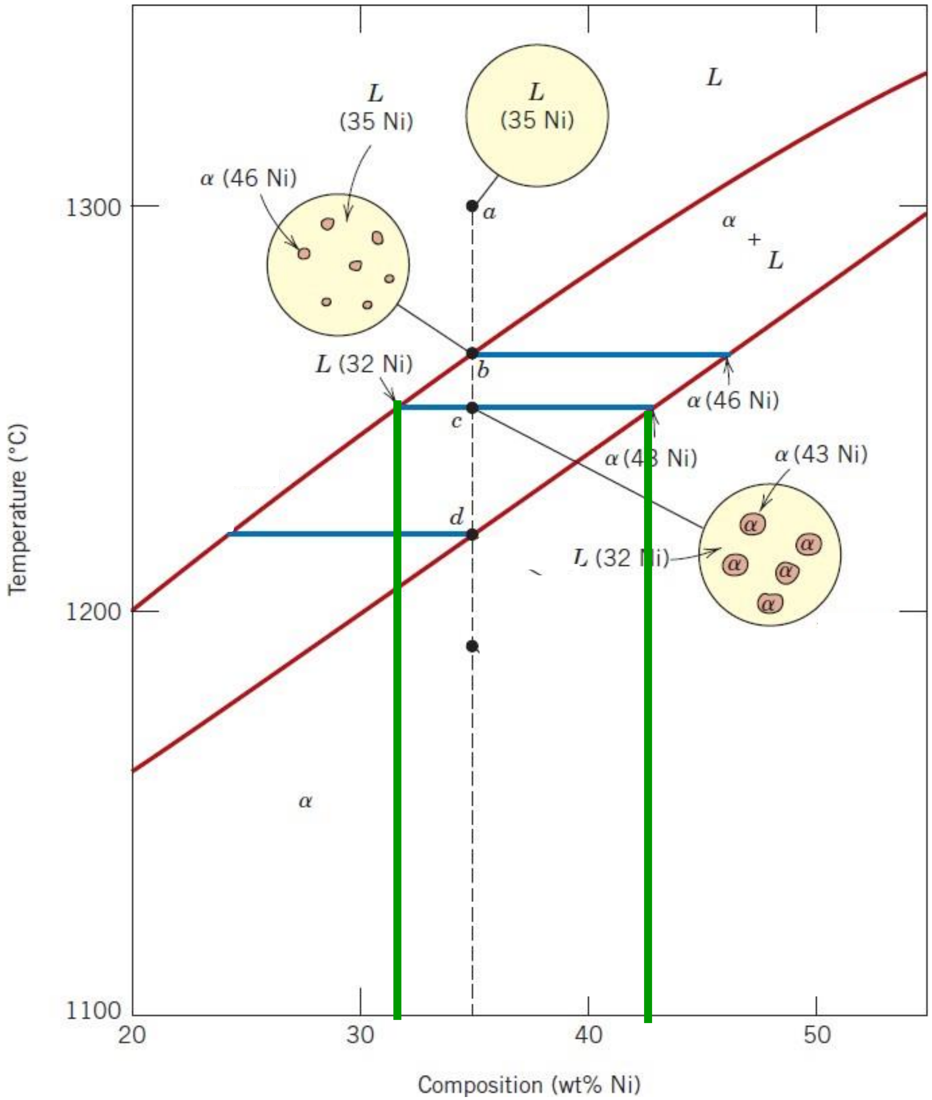


a:
Liquid phase: $35\ wt\%$ Ni and $65\ wt\%$ Cu

b:
α phase: $46\ wt\%$ Ni and $54\ wt\%$ Cu +
Liquid phase: $35\ wt\%$ Ni and $65\ wt\%$ Cu

Equilibrium Cooling

Slow Cooling of Copper-Nickel Alloy



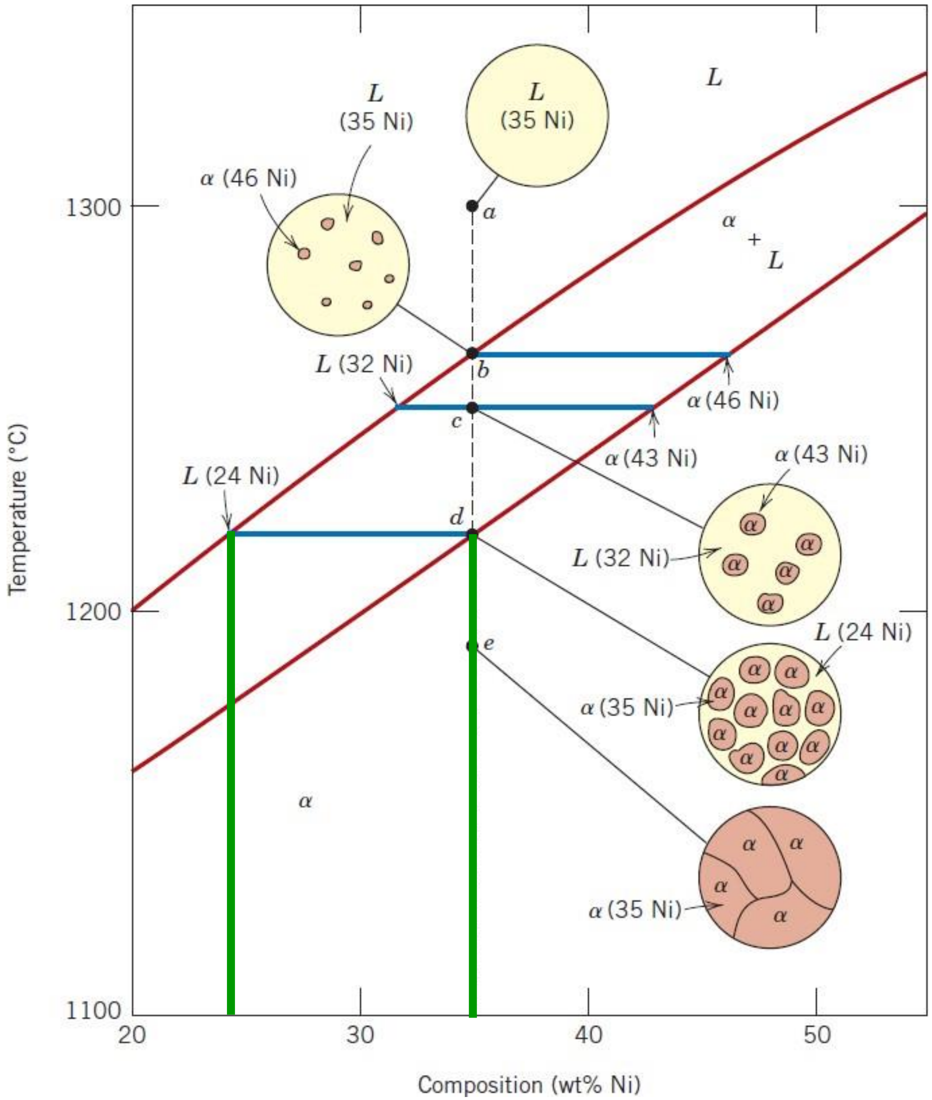
a:
Liquid phase: $35\ wt\%$ Ni and $65\ wt\%$ Cu

b:
α phase: $46\ wt\%$ Ni and $54\ wt\%$ Cu +
Liquid phase: $35\ wt\%$ Ni and $65\ wt\%$ Cu

c:
α phase: $43\ wt\%$ Ni and $57\ wt\%$ Cu +
Liquid phase: $32\ wt\%$ Ni and $68\ wt\%$ Cu

Equilibrium Cooling

Slow Cooling of Copper-Nickel Alloy



a:
Liquid phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu

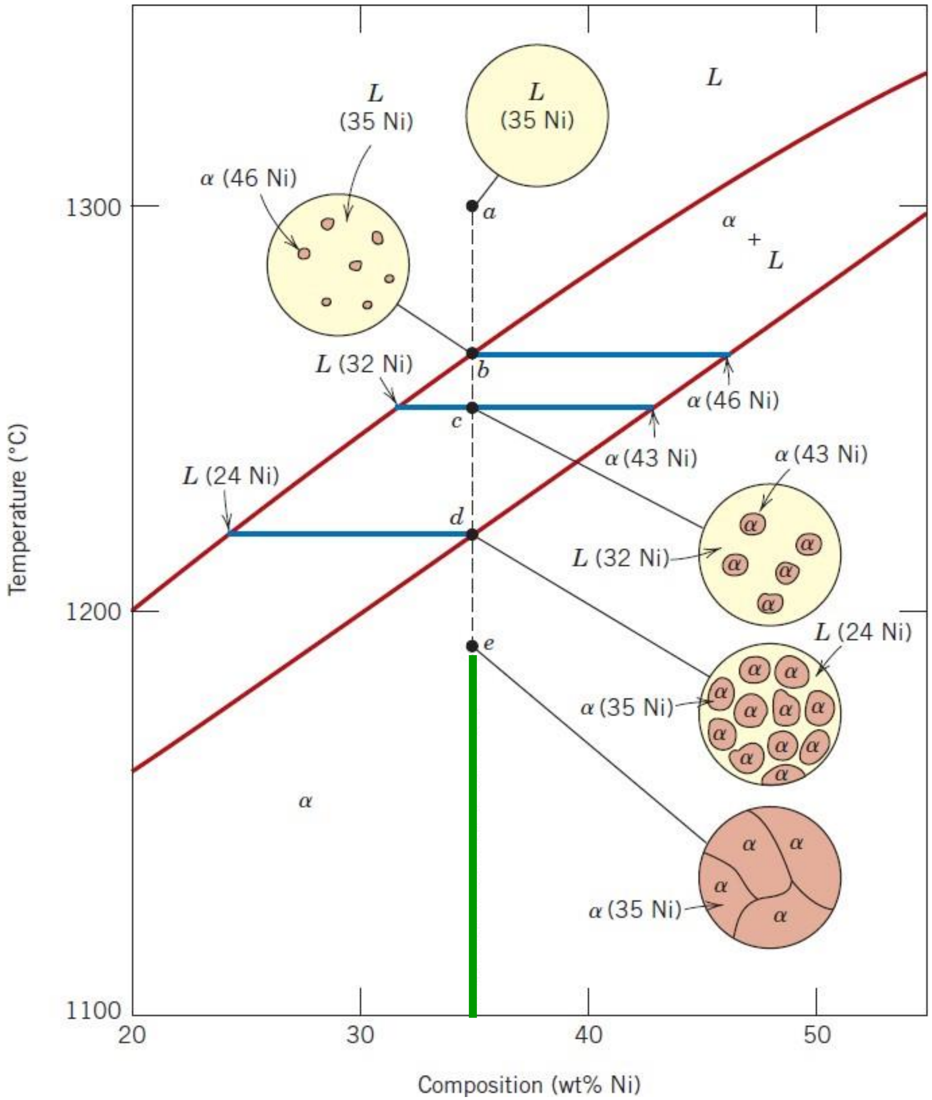
b:
α phase: 46 $wt\%$ Ni and 54 $wt\%$ Cu +
Liquid phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu

c:
α phase: 43 $wt\%$ Ni and 57 $wt\%$ Cu +
Liquid phase: 32 $wt\%$ Ni and 68 $wt\%$ Cu

d:
α phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu +
Liquid phase: 24 $wt\%$ Ni and 76 $wt\%$ Cu

Equilibrium Cooling

Slow Cooling of Copper-Nickel Alloy



a:
Liquid phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu

b:
α phase: 46 $wt\%$ Ni and 54 $wt\%$ Cu +
Liquid phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu

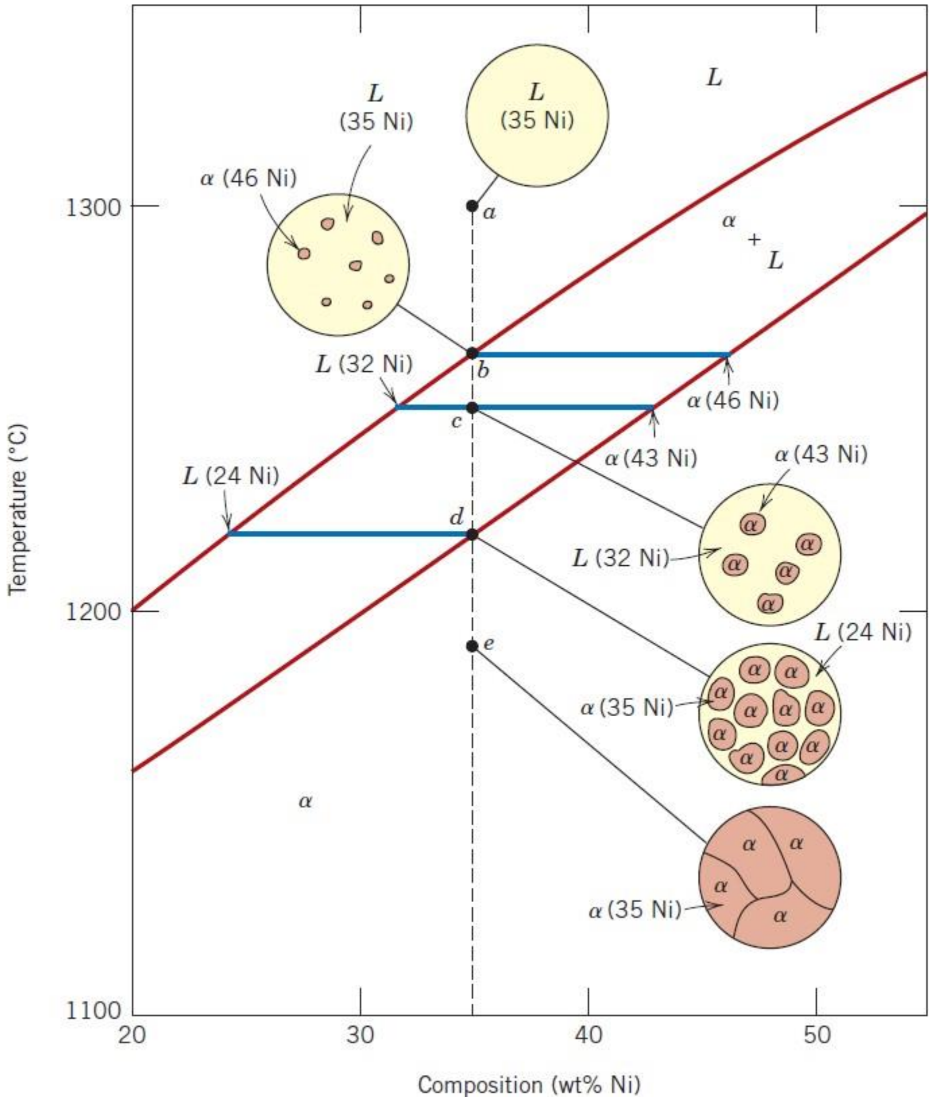
c:
α phase: 43 $wt\%$ Ni and 57 $wt\%$ Cu +
Liquid phase: 32 $wt\%$ Ni and 68 $wt\%$ Cu

d:
α phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu +
Liquid phase: 24 $wt\%$ Ni and 76 $wt\%$ Cu

e:
α phase: 35 $wt\%$ Ni and 65 $wt\%$ Cu +

Equilibrium Cooling

Slow Cooling of Copper-Nickel Alloy

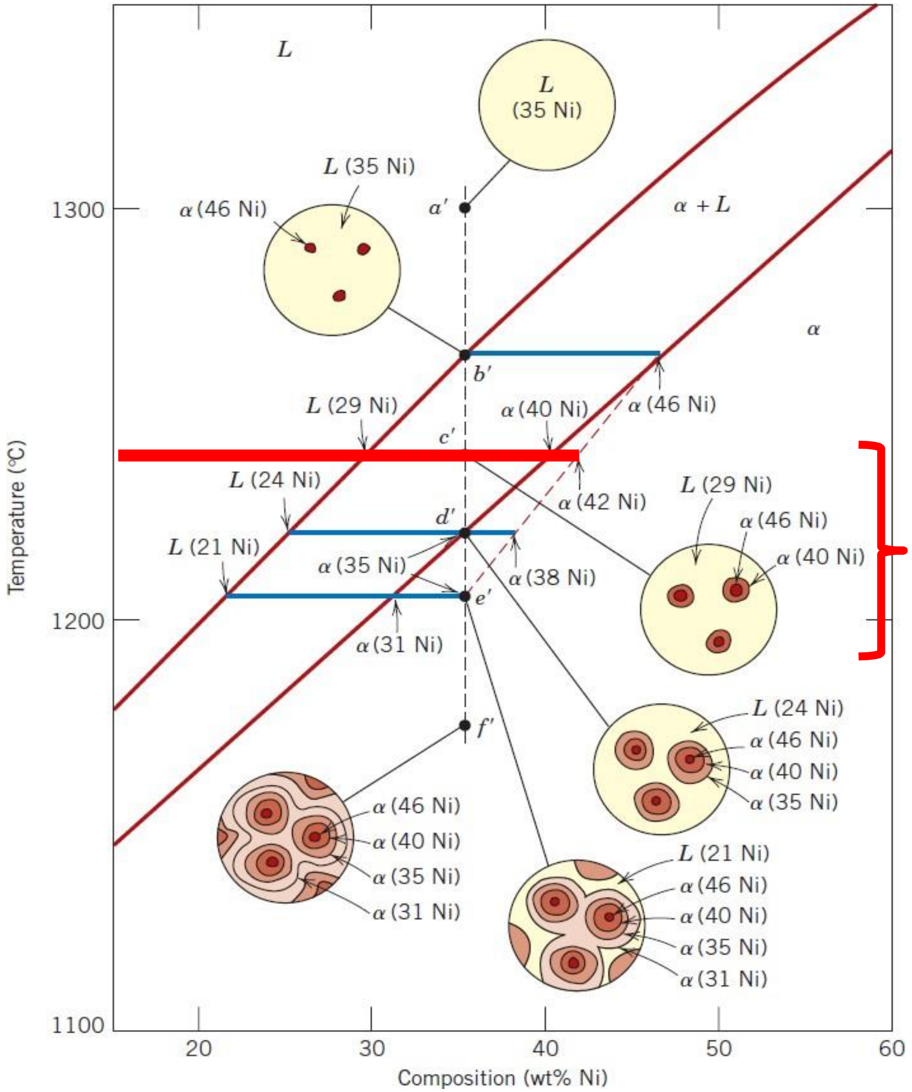


- For slow cooling, the diffusion rates, defined by the magnitudes of the diffusion coefficients, decrease with lower temperature.
- In practical solidification situations, the cooling rates are much too rapid to allow these compositional readjustments and maintenance of equilibrium.
- Consequently, the microstructures upon rapid cooling will be different than those of equilibrium cooling.

Non-equilibrium Cooling

Non-Equilibrium Cooling

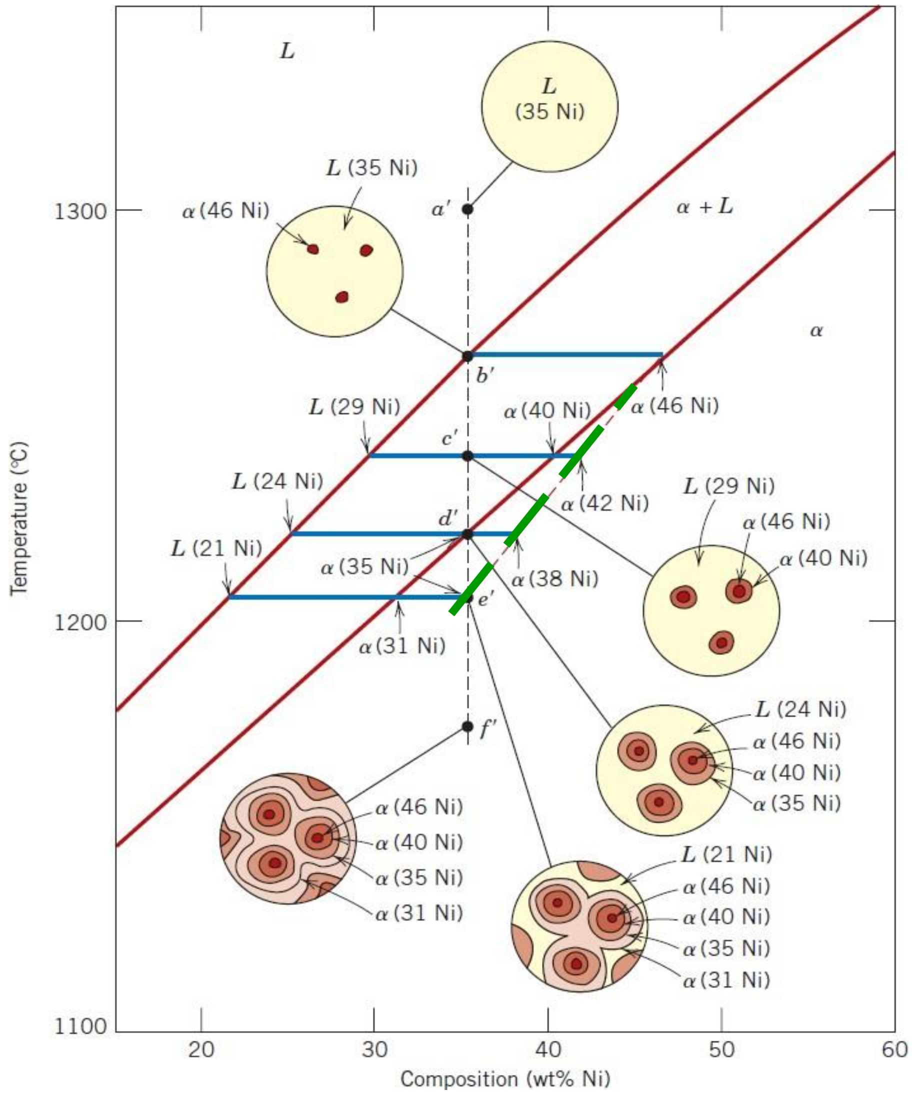
Rapid Cooling of Copper-Nickel Alloy ($35\ \%wt$ Ni and $65\ \%wt$ Cu)



- The diffusion rates in the liquid phase are sufficiently rapid such that equilibrium is maintained in the liquid phase.
- Upon further cooling (at about 1240°C), the liquid composition has shifted to $29\ wt\%$ Ni and $71\ wt\%\ Cu$ and the composition of the phase that solidifies is $40\ wt\%$ Ni and $60\ wt\%$ Cu.
- However, since diffusion in the solid phase is relatively slow, the phase that formed at point *b'* has not changed composition, it is still about $46\ wt\%$ Ni and the composition of the grains has continuously changed with radial position, from $46\ wt\%$ Ni at grain centers to 40 wt% Ni at the outer grain perimeters.
- Based on lever rule, a greater proportion of liquid is present for these non-equilibrium conditions than for equilibrium cooling.

Non-Equilibrium Cooling

Rapid Cooling of Copper-Nickel Alloy (35 $\%wt$ Ni and 65 $\%wt$ Cu)

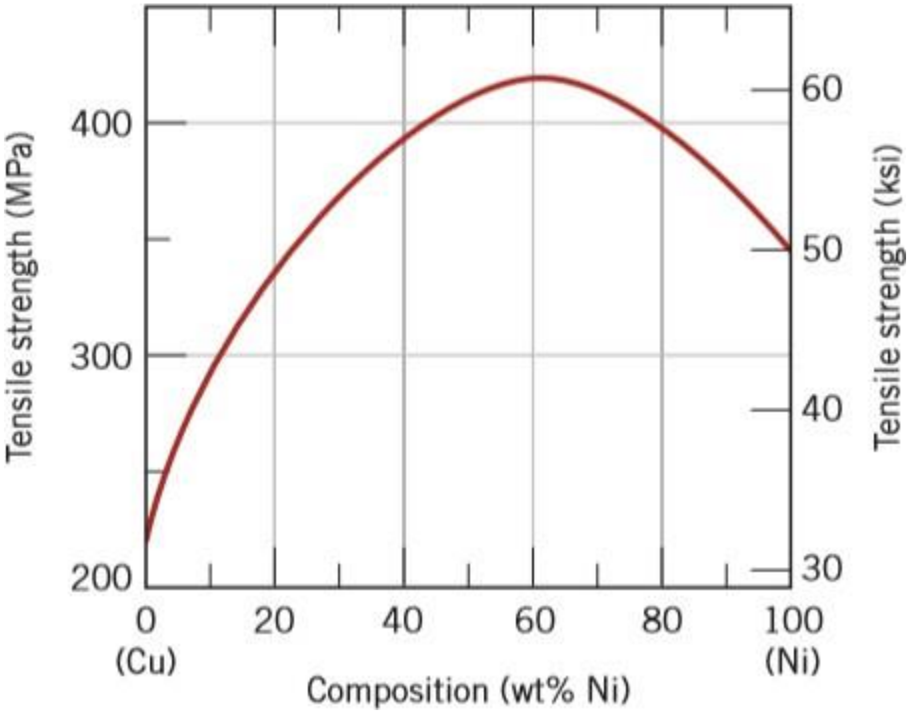


- For **non-equilibrium solidification** phenomenon, the **solidus line** on the phase diagram has been **shifted** to higher Ni contents ($40\ wt\%$ Ni to $42\ wt\%$ Ni)

- There is **no change** of the **liquidus line** since it is assumed that the equilibrium is maintained in the liquid phase during cooling due to sufficiently rapid diffusion rates.

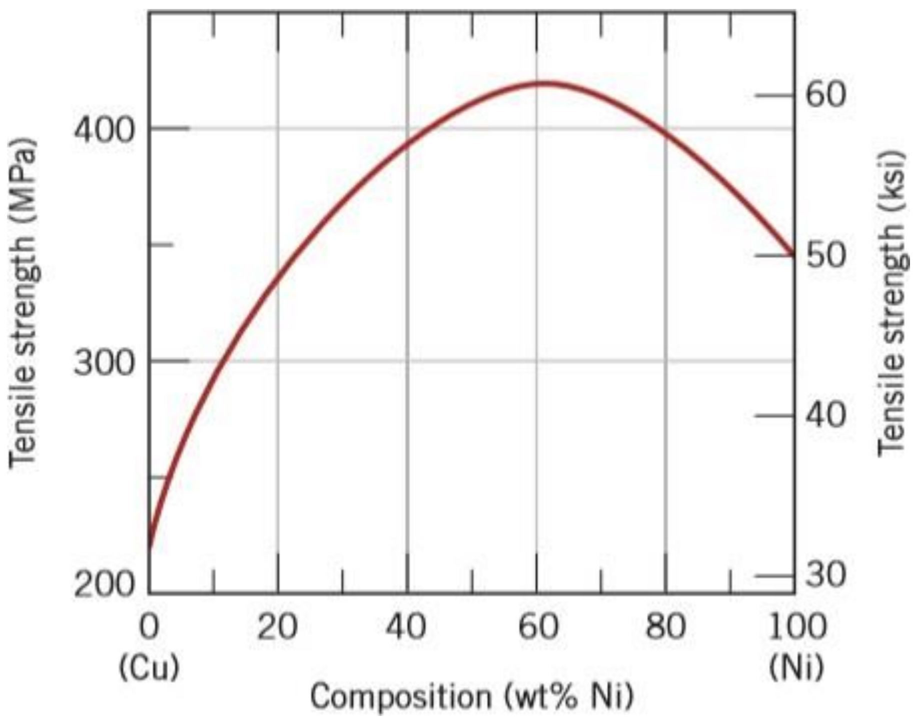
Mechanical Properties of Isomorphous Alloys

Tensile Strength versus Composition for Copper-Nickel Alloy

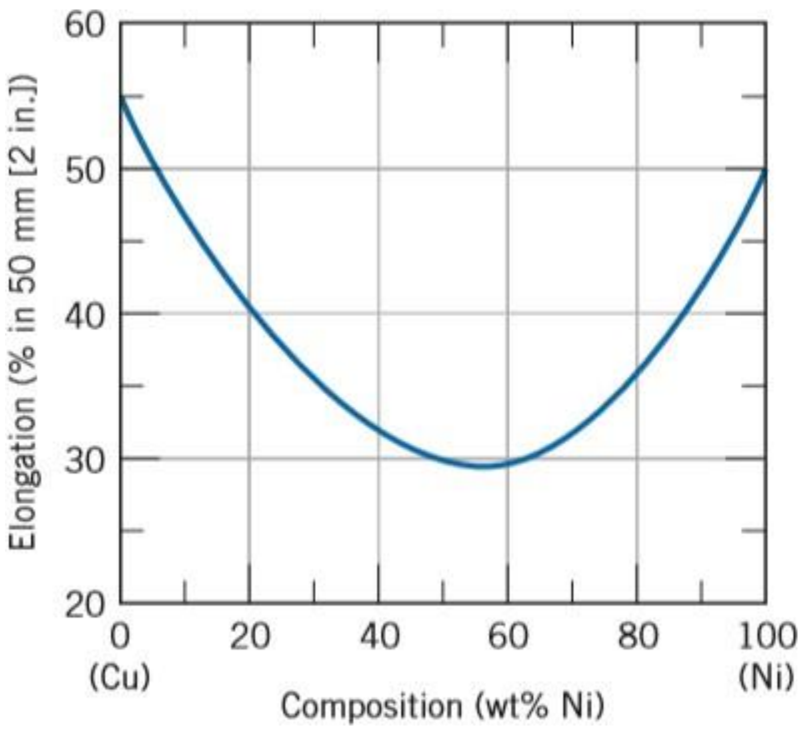


- Each component will experience **solid-solution strengthening** ,or in other words an increase in strength and hardness by additions of the other component.

- The **tensile strength** increases as the composition of nickel increases then at some intermediate composition, the tensile strength passes through a maximum and decreases again.



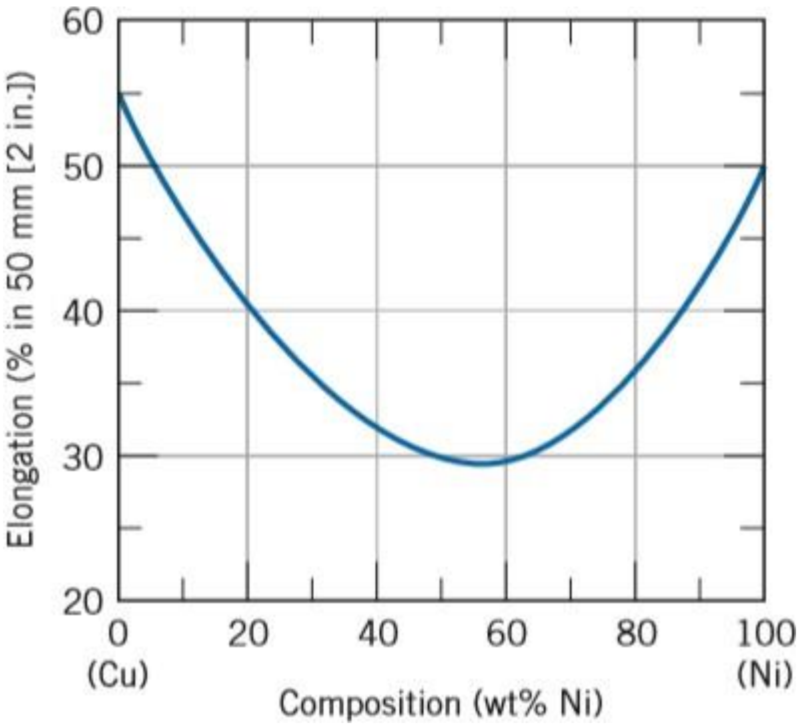
Tensile strength (MPa)
200
300
400
Tensile strength (ksi)
30
40
50
60
0
20
40
60
80
100
(Cu)
(Ni)
Composition (wt% Ni)



Elongation (% in 50 mm [2 in.])
20
30
40
50
60
0
20
40
60
80
100
(Cu)
(Ni)
Composition (wt% Ni)

Mechanical Properties of Isomorphous Alloys

Ductility (%EL) versus Composition for Copper-Nickel Alloy



- The **ductility** decreases as the composition of nickel increases then at some intermediate composition, the ductility passes through a minimum and increases again.

Gibbs Phase Rule

- Gibb's phase rule allows to determine the number of degrees of freedom (F) or variance of the alloy as a function of number of components and phases.

$$P + F = C + N$$

P: Number of phases present

C: Number of components on the system

N: Number of non-compositional variables

Example: Gibbs Phase Rule

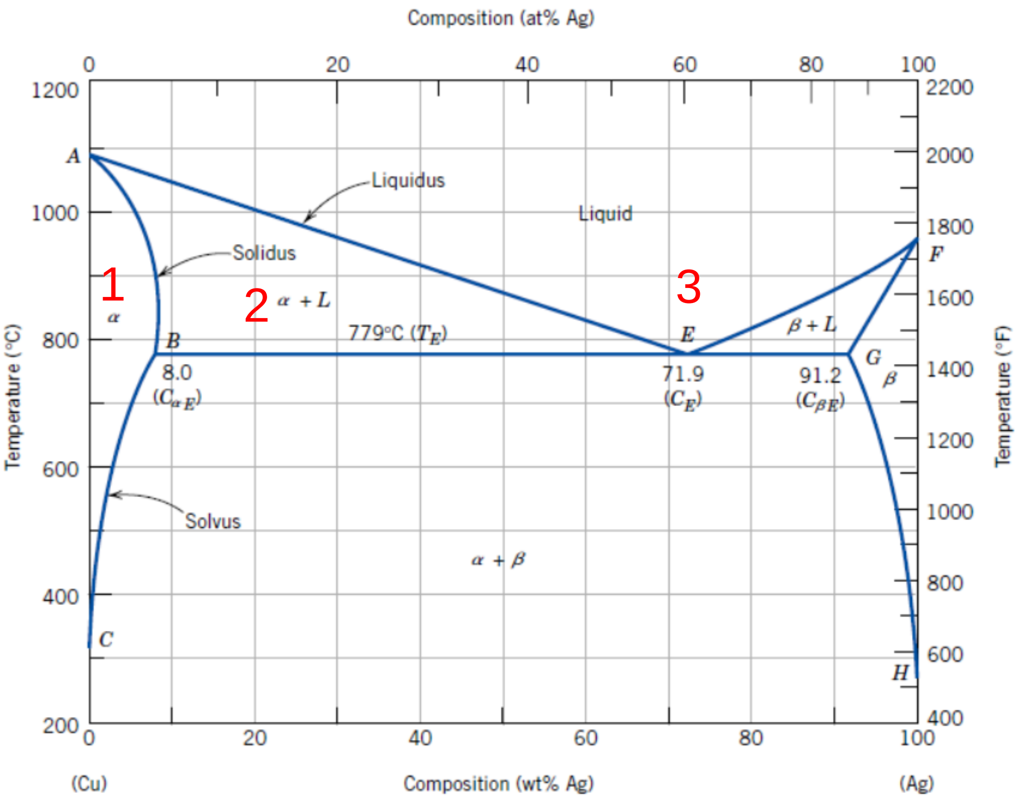


Figure 9.7 The copper–silver phase diagram. [Adapted from *Binary Alloy Phase Diagrams,* 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

$$P + F = C + N$$

Point 1:

Point 2:

Point 3:

Example: Gibbs Phase Rule

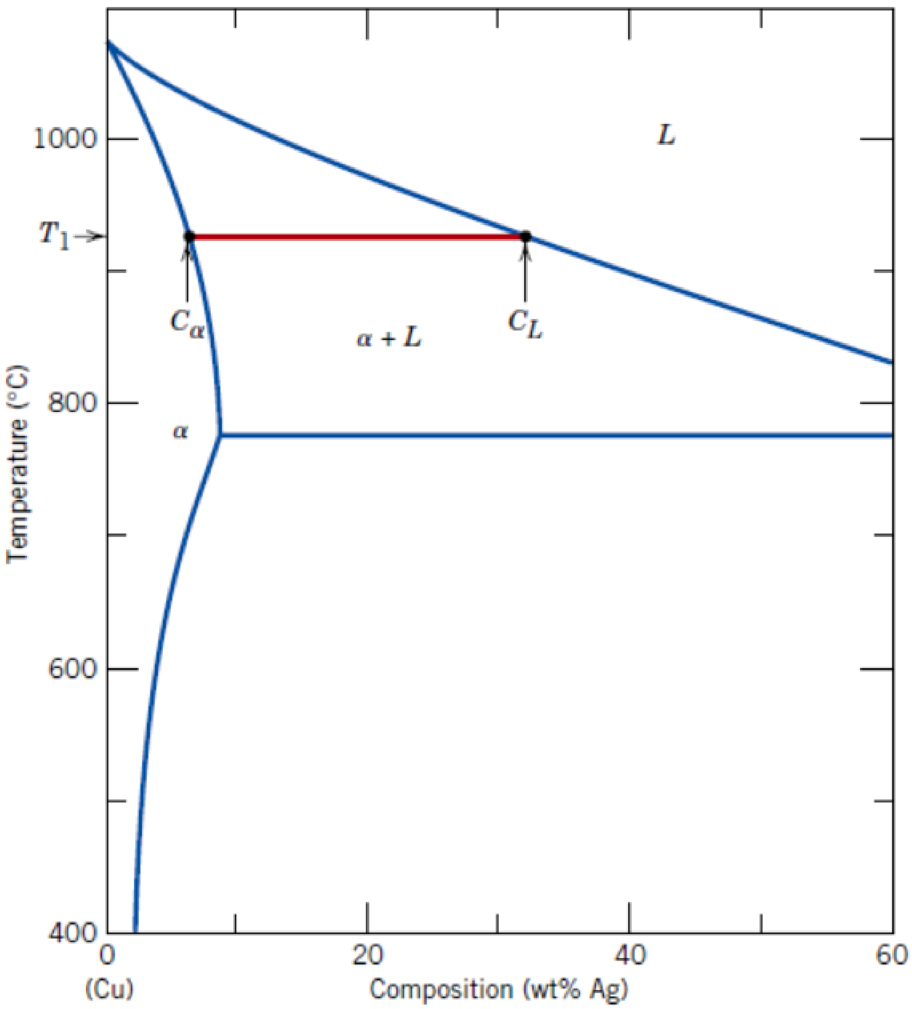
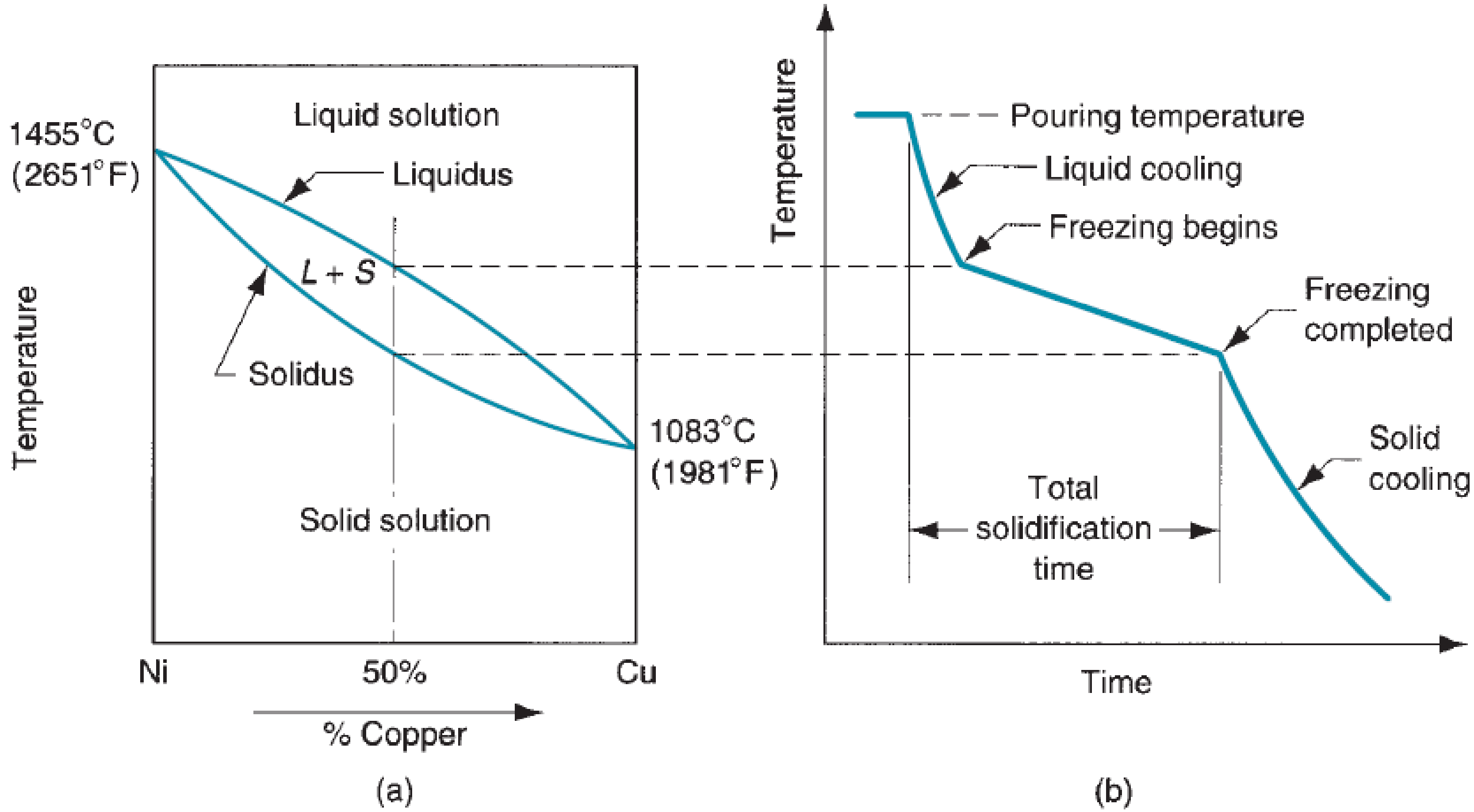


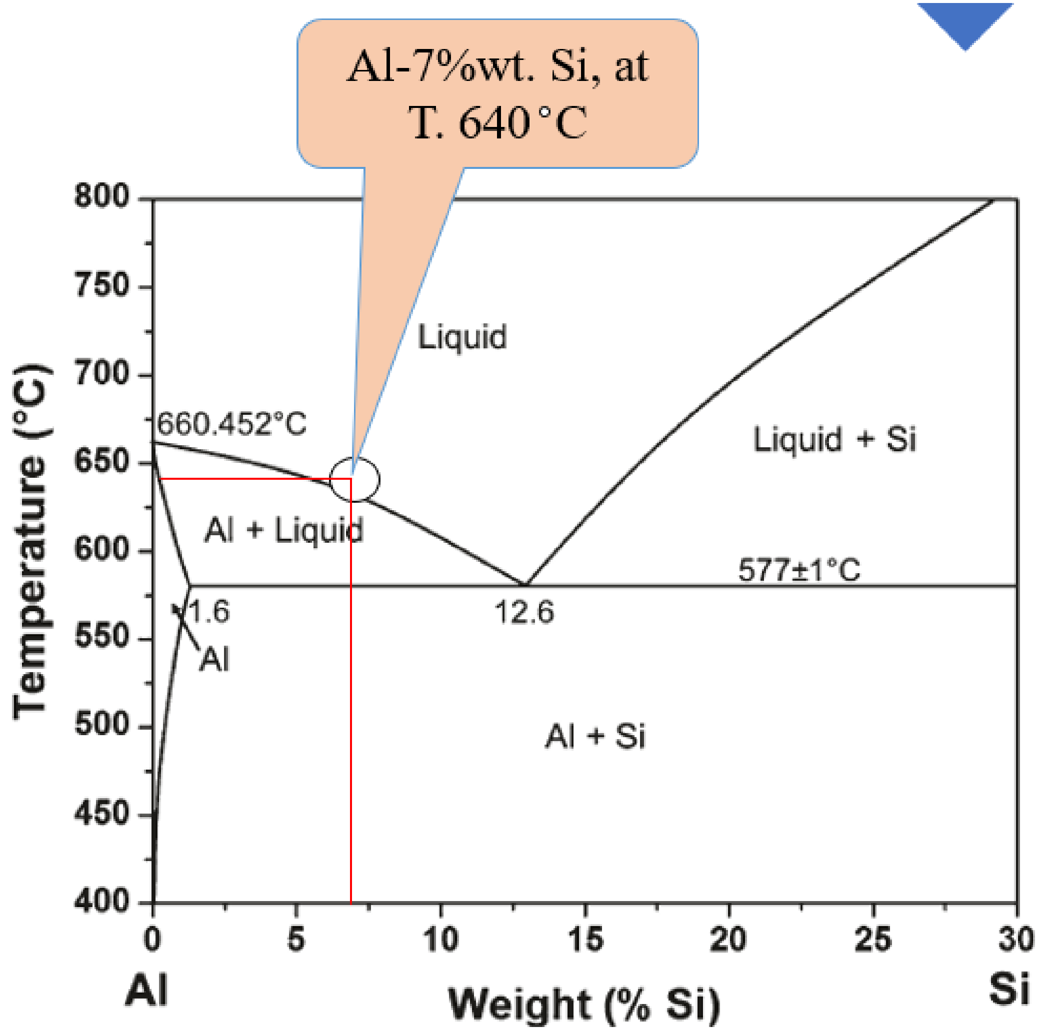
Figure 9.23 Enlarged copper-rich section of the Cu–Ag phase diagram in which the Gibbs phase rule for the coexistence of two phases (α and L) is demonstrated. Once the composition of either phase (C_α or C_L) or the temperature (T_1) is specified, values for the two remaining parameters are established by construction of the appropriate tie line.

Related Phase Diagram and Cooling Curve



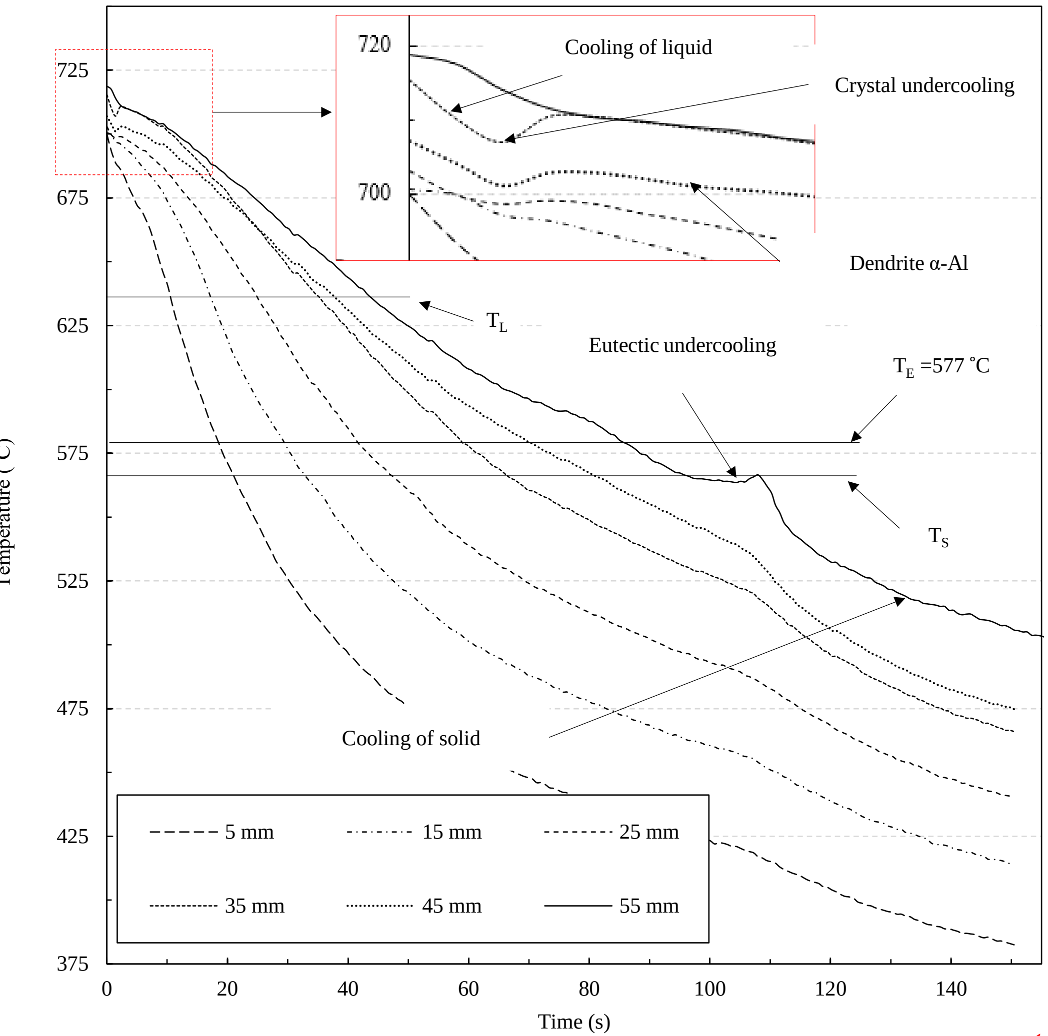
(a) Phase diagram for a copper–nickel alloy system and (b) associated cooling curve for a 50%Ni–50%Cu composition during casting.

Phase Diagram ocf Al-Si Alloy

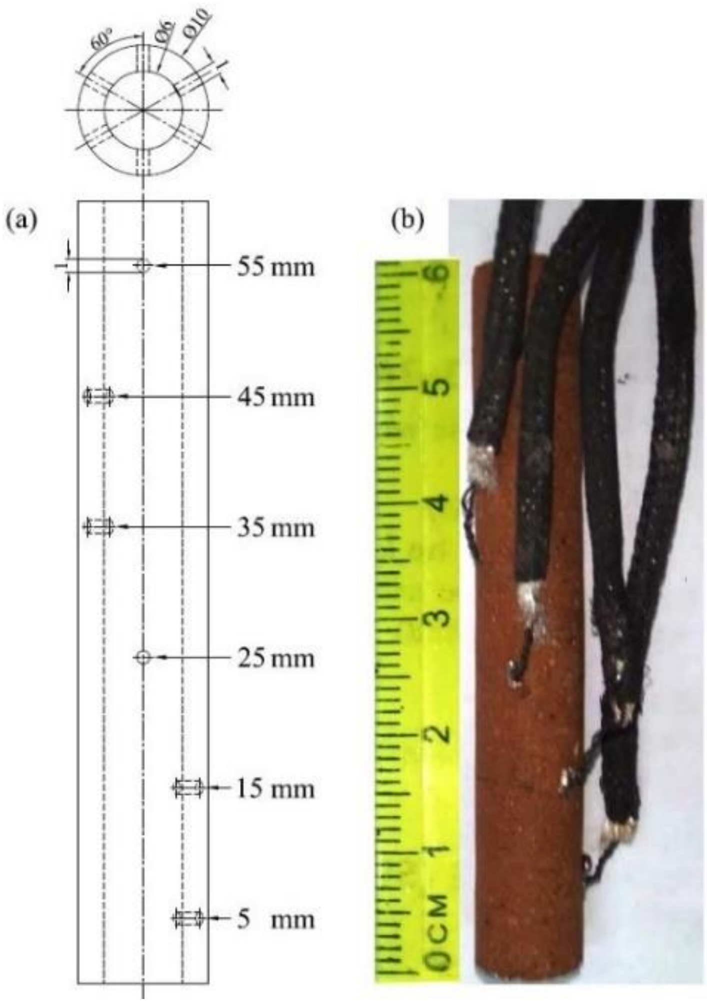


Binary Al-Si phase diagram

Example: Cooling Curves

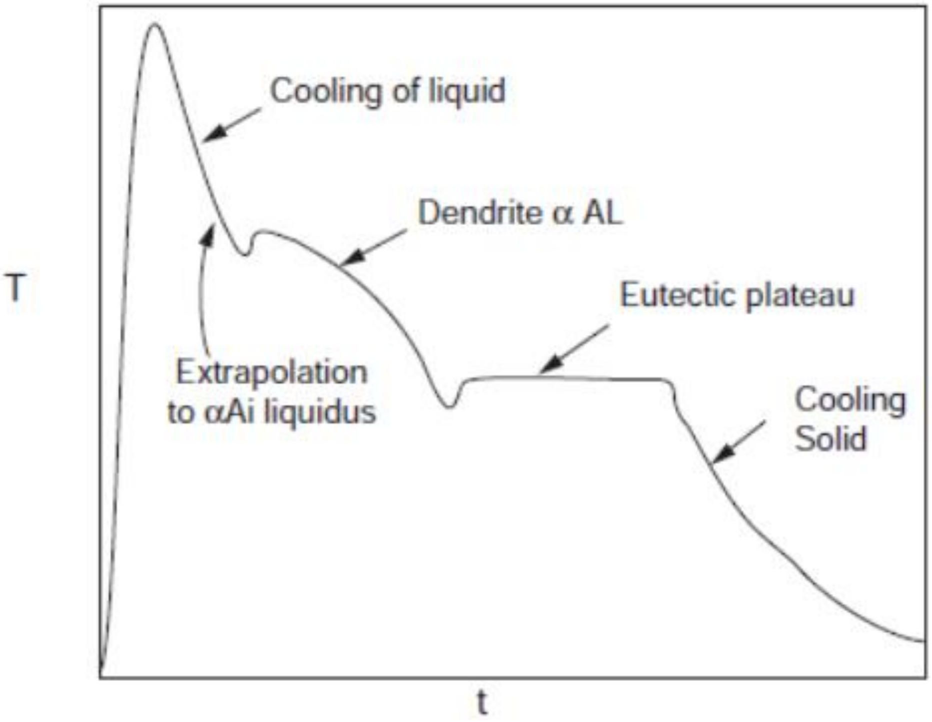


Cooling curve of directionally solidified in during solidification

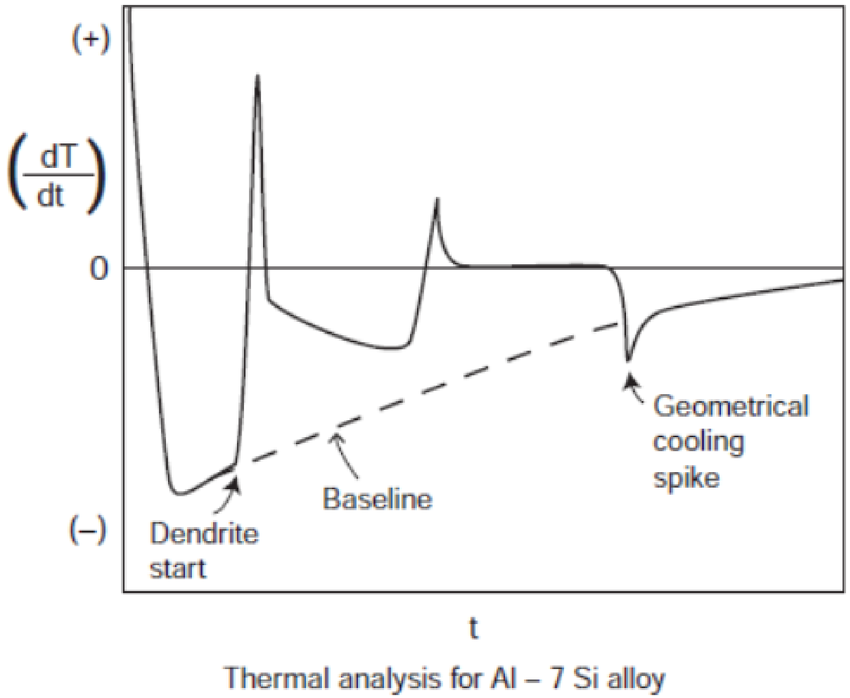


(a) Engineering drawing of mold and (b) the sample formed in mold and mounted with thermocouples

Example: Cooling Curves

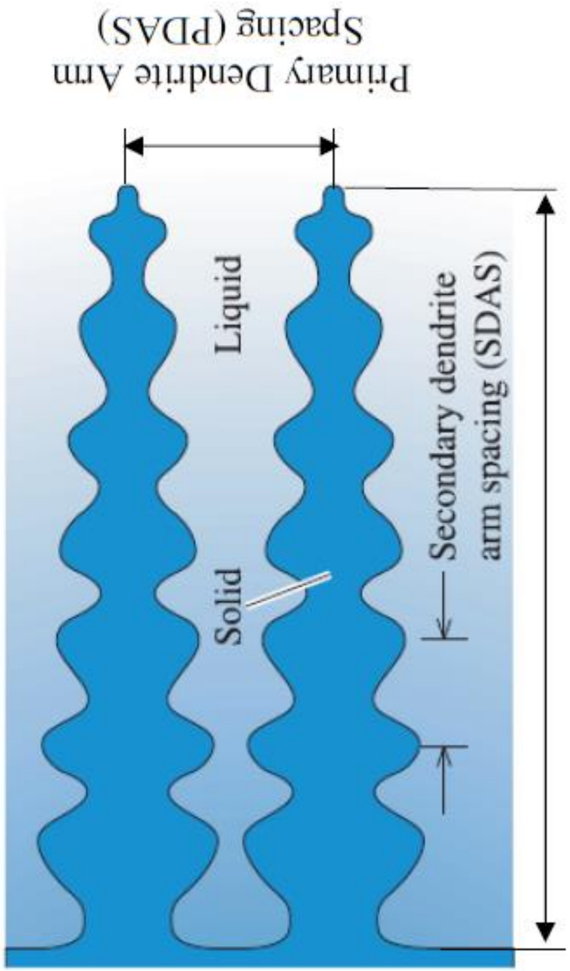


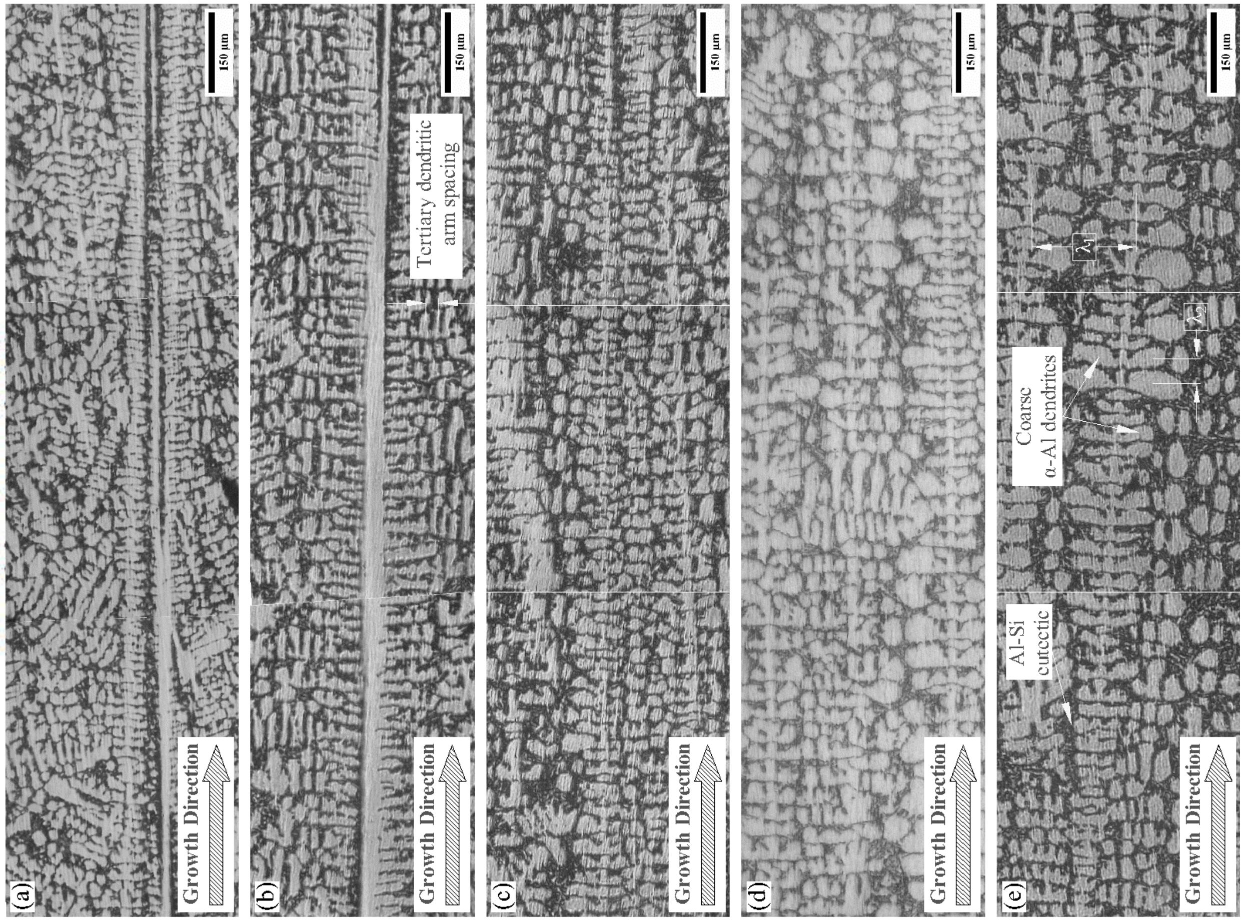
Cooling Curve of Al-Si 7wt% of Si



First Derivative of Cooling Curve of Al-Si 7wt% of Si

Example: Cooling Curves





Microstructure Develop During Cooling of Al-Si 7wt% Alloy, Thank you for your attention!